

Learning objectives

In this lesson students will learn to:

- Compare the capacity, speed and portability of magnetic, optical and solid-state storage devices
- Select an appropriate type of storage for a particular purpose
- Construct an expression to calculate data storage requirements.

For more details on this topic and additional student activities see Topics 2 and 3 of the student book.

Recap

Last lesson you learned about the three types of storage: magnetic, optical, and solid state.

You learned about how they work and looked at some of the basic pros and cons of each.

Today, you are going to examine in detail the differences between the three types and discuss why you would choose one over the others in specific situations.

Things to consider

You need to think about the criteria you will use for comparison, before you actually do the work.

You need to consider the following aspects of each type of data storage:

- capacity – how much data can the device store?
- portability – can the device be moved around without damaging it or the data it contains?
- speed – how fast can data be read from and written to the device?

Choosing a device for a scenario

Now that you know a lot about the different types of data storage technology, you need to be able to choose the most appropriate type to use for a given situation.

For example:

Your school has a small handheld video camera that a group of students need to use for Geography fieldwork. What would the most appropriate storage type to use with the camera and why?

Answer:

A solid state memory card would be most appropriate. It would have a large enough capacity for video files and could be added to with additional memory cards. Such cards have a high write speed, which will support more data being read to the storage medium, and allow students to film in high quality. They are also highly portable – memory cards are small and have no moving parts that can be broken.

Capacity

Until recently, standard decimal prefixes – kilo, mega, giga, etc. – have been used to represent binary multiples. This has caused some confusion.

To address this, the International Electrotechnical Commission (IEC) has produced a set of binary prefixes to represent binary multiples.

These are the units of measurement you are expected to use for data storage and memory capacity.

Unit	Abbreviation	Bytes	Equivalent to
bit			1 bit
nibble			4 bits
byte		2^0 bytes	8 bits or 2 nibbles
kibibyte	KiB	2^{10} bytes	1024 bytes
mebibyte	MiB	2^{20} bytes	1024 kibibytes
gibibyte	GiB	2^{30} bytes	1024 mebibytes
tebibyte	TiB	2^{40} bytes	1024 gibibytes

Binary multiples

Conversion between the units is straightforward.

Example:

A hard disk has a storage capacity of 1.5 TiB

Express this in:

- mebibytes
- Kibibytes

$1\text{TiB} = 1024\text{ MiB}$ so $1.5\text{ TiB} = 1024 * 1.5\text{ MiB}$

$1\text{MiB} = 1024\text{ KiB}$ so $1.5\text{ TiB} = 1024 * 1024 * 1.5\text{ KiB}$.

Exam note:

You are **not** allowed a calculator in the exam.

This means you are not expected to actually calculate the answers to questions like this.

Instead you should construct an **expression** to show **how** the answer would be calculated.

Wrap up: you have learned how to...

- ✓ Compare the capacity, speed and portability of magnetic, optical and solid-state storage devices.
 - Capacity is measured in mebibytes, gibibytes and tebibytes.
 - Speed refers to how fast data can be written to and read from a device.
 - Portability means how much movement the device can be subjected to without causing damage to the device or the data it contains.

and you have learned how to...

- ✓ Select an appropriate type of storage for a particular purpose.
Use the capacity, speed and portability to decide which device type would be most suitable.
- ✓ Construct an expression to calculate data storage requirements.
 - For exams, you don't need to calculate a full answer, just construct an expression to show how we would complete the calculation.